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SQL Exercise: Aggregate Functions & Grouping

Questions and Answers

1. COUNT () Function

Write a SQL query to find the **total number of employees** in the company.

Answer:

```
SELECT COUNT (*) AS total_employees

FROM employees_db;
```

total_employees
10

2. SUM () Function

Write a SQL query to find the **total salary** paid to all employees in the **IT department**.

Answer:

```
SELECT SUM (salary) AS total_it_salary

FROM employees_db

WHERE department = 'IT';
```

total_IT_salary
220 000

3. AVG () Function

Write a SQL query to calculate the **average salary** of employees in the **HR department**.

Answer:

```
SELECT AVG (salary) AS avg_hr_salary

FROM employees_db

WHERE department = 'HR';
```

avg_hr_salary
49 500

4. MIN () and MAX () Functions

Write a SQL query to find the **highest and lowest salary** in the company.

Answer:

```
SELECT MIN (salary) AS lowest salary,
        MAX (salary) AS highest_salary
FROM employees_db;
```

lowest_salary	highest_salary
48 000	62 000

5. GROUP BY Statement

Write a SQL query to **group employees by department** and display the **total salary paid in each department**.

Answer:

```
SELECT department,
        SUM (salary) AS total_salary
FROM employees_db
GROUP BY department;
```

department	total_salary
IT	220 000
HR	99 000
Finance	119 000
Marketing	105 000

6. GROUP BY and COUNT ()

Write a SQL query to count how many employees work in each **city**.

Answer:

```
SELECT city, COUNT (*) AS employee_count
```

```
FROM employees_db

GROUP BY city;
```

city	employee_count
New York	2
Chicago	3
Los Angeles	2
San Francisco	2
Houston	1

7. GROUP BY and ORDER BY

Write a SQL query to group employees by **department**, calculate the **average salary** in each department, and **order the results in descending order** of average salary.

Answer:

```
SELECT department, AVG (salary) AS avg_salary

FROM employees_db

GROUP BY department

ORDER BY avg_salary DESC;
```

department	avg_salary
Finance	59 500
IT	55 000
Marketing	52 500
HR	49 500

8. HAVING Clause

Write a SQL query to find **departments where the total salary paid exceeds 100,000**. (Use GROUP BY and HAVING).

Answer:

```
SELECT department, SUM (salary) AS total_salary

FROM employees_db

GROUP BY department

HAVING SUM (salary) > 100000;
```

department	total_salary
IT	220 000
Finance	119 000
Marketing	105 000

9. Combining GROUP BY, HAVING, and ORDER BY

Write a SQL query to **list cities where more than one employee works**, **ordered by the number of employees in descending order**.

Answer:

```
SELECT city, COUNT (*) AS employee_count

FROM employees_db

GROUP BY city

HAVING COUNT (*) > 1

ORDER BY employee_count DESC;
```

city	employee_count
Chicago	3
New York	2
Los Angeles	2
San Francisco	2

10. Combining Aggregate Functions

Write a SQL query to find the **department with the highest average salary**.

Answer:

```
SELECT department, AVG (salary) AS avg_salary

FROM employees_db

GROUP BY department

ORDER BY avg_salary DESC

LIMIT 1;
```

department	avg_salary
Finance	59 500